



The OPEN SOURCE CHIP MOVEMENT Has Just Begun



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In October 2018, India saw the launch of the first indigenously built chip that can change the Indian consumer and industrial electronics landscape. Baishakhi Dutta catches up with G.S. Madhusudan, senior project advisor at IIT-Madras, to talk about their design strategies for SHAKTI microprocessor and how it can serve the electronics community at large

Q. What's the technology behind SHAKTI and its applications?

A. SHAKTI was started not just as a research project but to also develop production-ready processor cores, which are in competition with equivalent processors and are better in areas like security and functional safety. We will be competitive with equivalent ARM cores, better in some respects and lagging in others. We will know more once SHAKTI gets wider adoption.

The research part of SHAKTI focuses on areas like supercomputers, neuromorphic computing and security techniques that use it as a base for research on publicly available platforms. This enables seamless translation into products and easier collaboration with other researchers.

SHAKTI aims to develop six families of processors, for everything from what goes inside door locks to supercomputers. For this kind of range, you cannot have a single design. In SHAKTI, E class is the lowest, then comes C class (for low-end cellphones, industrial controllers), I class, S class (for servers), H class (for supercomputers), F class (for functional safety processors for aerospace/auto/medical applications) and finally T class (for high-security applications).

Q. How do you ensure that the chips are at par with international standards?

A. When we evaluate chips, there are basically three broad parameters: performance, power consumption and size. The instruction set is an international standard called RISC-V. The RISC-V consortium, of which IIT-M is a founding member, comprises experts from around the world, who have designed RISC-V ISA to be an extensible platform for building next-generation processors.

Q. What certifications and quality assurance testing has the chip gone

through to ensure maximum usage?

A. SHAKTI has an open source-based test methodology and a comprehensive set of test suites. This is at par with standard industry practices. We are also working on formal verification techniques.

After the chip is fabricated, the foundry or entity that manufactures the chip does chip-level qualification testing. Then, we re-run our standards tests and some application tests to check for correctness, stability and performance.

Q. Is the chip ready for such emerging technologies as 5G, machine learning (ML) or artificial intelligence (AI)?

A. IIT-M develops the basic cores and proves them on test chips. The startups then take these cores and build ASICs for the above-mentioned applications.

Q. How competitive is the chip in terms of pricing?

A. In terms of development and IP cost, it is cheaper. Ultimately, cost of the chip is tied to its size. In that respect, SHAKTI cores are very optimal and competitive with contemporary cores.

Q. How is open source playing a role in chip manufacturing?

A. IIT-M's computer science department is an advocate of open source and open standards for a long time, which led to the decision to co-found RISC-V consortium. The open source chip movement has just begun; how it will change the world remains to be seen.

Q. Who is RISC-V more beneficial to, chip designer or customer?

A. Both. For the designer, it allows easy design exploration, where an optimal design can be reached quicker.

For the customer, it offers a choice of vendors, since many vendors provide royalty-free RISC-V designs.

For example, a college kid with an idea can simply go to SHAKTI website, take Verilog code, compile it, test it out, make any required changes and within weeks create a new design.

Q. What kind of application areas is RISC-V platform best suited for?

A. Pretty much any kind of design can be built with RISC-V since it is a general-purpose ISA.

Q. Will students' going to SHAKTI website help in its growth?

A. Increased student use of SHAKTI will increase its adoption and ensure availability of RISC-V-trained engineers. This will help strengthen the CPU design ecosystem in India. We will soon launch low-cost FPGA kits to help students experiment with SHAKTI.

Q. When will these kits come out?

A. The kits are ready, and should be available by September, once the software around them has been created.

Q. Any startups that have adopted SHAKTI?

A. InCore Semiconductors uses some of the SHAKTI IP. The startup focuses on AI/ML systems, security and fault-tolerant computing.

Q. Since the technology is open source, how do you monetise it?

A. Thales, one of the world's largest defence electronics company, is evaluating SHAKTI processors for aerospace applications, and has given a large grant to IIT-M to accelerate development. Many entities have started paying funds to IIT-M or are looking forward to collaborate to figure out how to use SHAKTI in their products.

Startups will provide commercial services on top of SHAKTI, or provide enhancements to SHAKTI. **EFY**